Pipeline for applying deep learning models to predict gene expression in diverse inverted repeat lacking clade legumes

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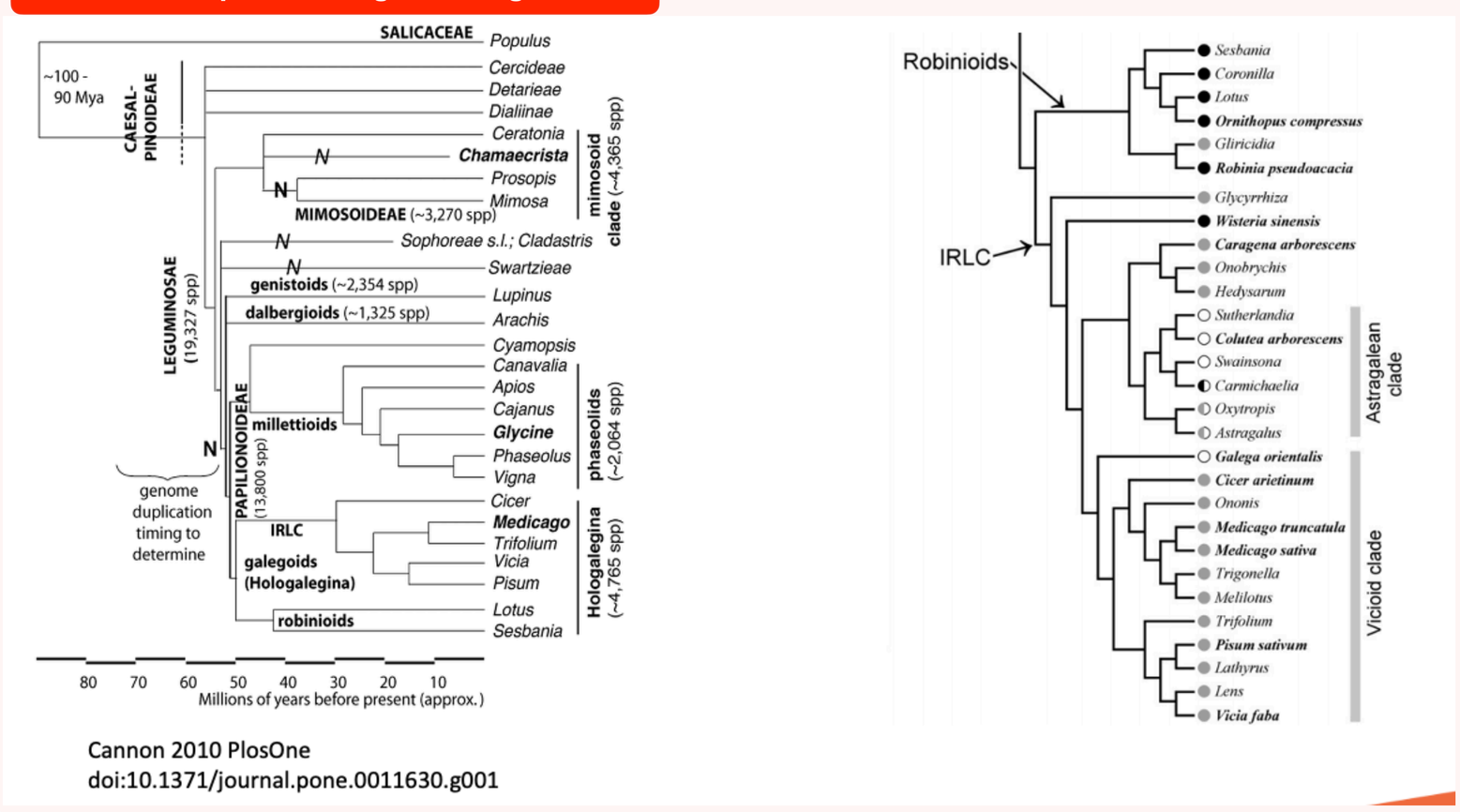
ABSTRACT

Ever increasing population growth and climate change are one of the greatest threats facing humanity in this century. State-of-art sequencing technologies have been developed to greatly facilitate the genomic and transcriptomic experiments, which present the opportunity to explore the correlation of gene expressions between species. However, there is growing need for faster and cheaper computational prediction comparing the time-consuming experimental tests. Here, the 'in-house' deep learning model - PlanTT were developed to predict gene expression from contrasting pairs of DNA sequence. We demonstrate the pipeline by applying this model tool on genome assemblies and matched transcriptomic experiments such as PRO-Seq and Quant-Seq. This includes a variety of inverted repeat lacking clade legumes such as, field pea (*Pisum sativum*), lentil species (*Lens sp.*), grass pea (*Lathyrus sativus*), faba bean (*Vicia faba*), chickpea (*Cicer arietinum*) and barrel medic (*Medicago truncatula*). We believe the workflow of the gene expression prediction will shed light on the understanding of environmental adaptation.

BACKGROUND

Inverted repeat lacking clade legumes



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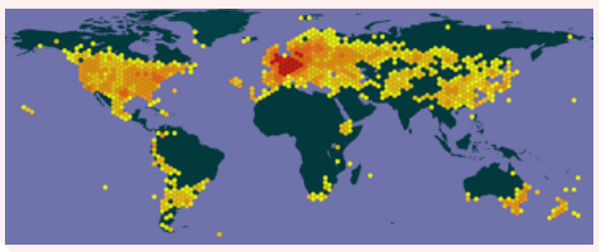
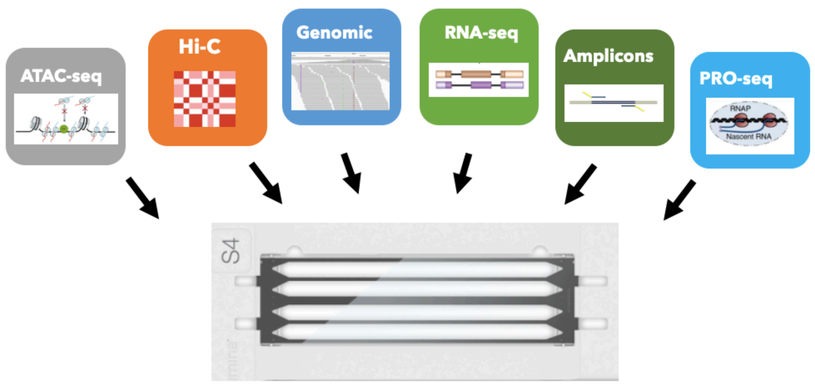
Sequencing strategies of the collected legumes

field pea (*Pisum sativum*), lentil species (*Lens sp.*), grass pea (*Lathyrus sativus*), faba bean (*Vicia faba*), chickpea (*Cicer arietinum*) and barrel medic (*Medicago truncatula*)





Allows combining many library types into a **single** high cost-efficiency lane

METHODS

Predicting gene expression from DNA sequence

Evolutionarily informed deep learning methods for predicting relative transcript abundance from DNA sequence

Jacob D. Washburn[a,1], Maria Katherine Mejia-Guerra[a], Guillaume Ramstein[a], Karl A. Kremling[a], Ravi Valluru[a], Edward S. Buckler[a,b,2], and Hai Wang[c,a,1,2]

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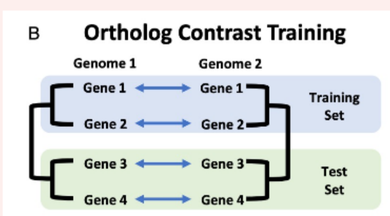
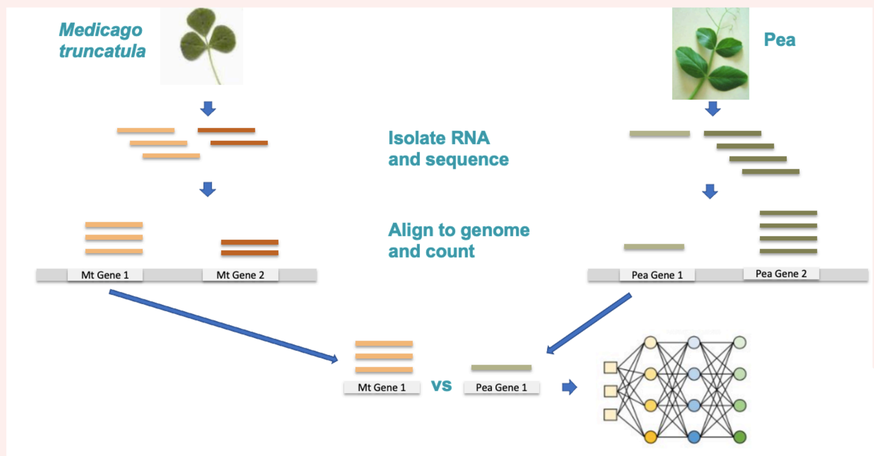
- Inspired by Washburn et al., we focused on the ortholog contract training.





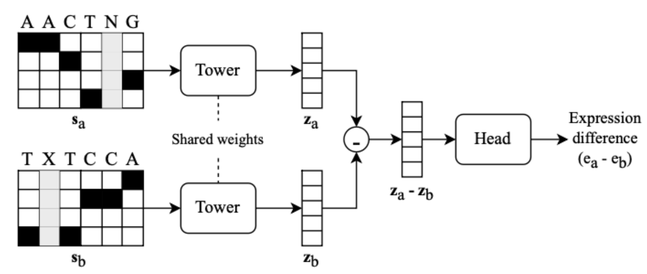
GOALS AND OBJECTIVES

- Consistent (pre)-processing of multiple genomes so they can be integrated into the PlanTT model pipeline
- Syntenic ortholog identification for multiple legume species by using OrthoFinder and MCscanX
- Providing transcript level and transcription labels for orthtolog pairs using custom scripts
- Validate gene expression prediction workflow by deploying AMII workflows on NRC servers
- Establishing methods for visualizing meta gene information, i.e., set up the pipeline with visualization tools (e.g., deeptools 2.0) on bigwig files.

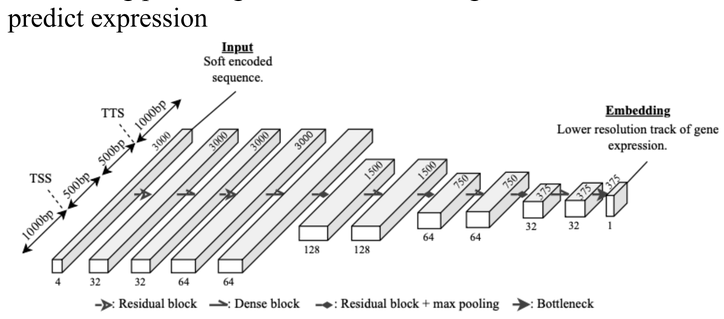
RESULTS

To explore optimization of gene expression models, we evaluated different internal normalization approaches to enable count-based rather than rank-based gene expression prediction.

Bioinformatics pipelines



Contrasting pairs of genes used for training neural networks that predict expression



Architecture of the Convolutional Neural Network (CNN) tower

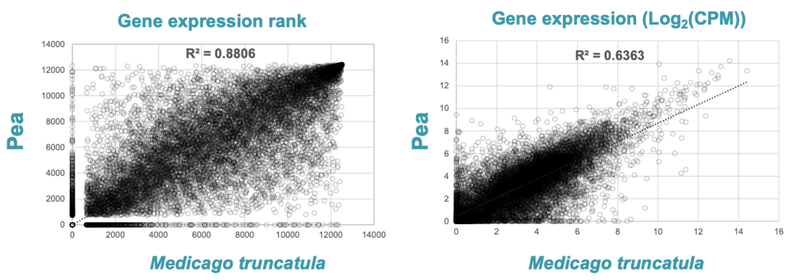
Performance of contract models

- **Reference genotypes from 4 species:**
  - *Pisum sativum* (Cameor)
  - *Medicago truncatula* (A17)
  - *Vicia faba* (Hedin/2)
  - *Lathyrus sativus* (LS007)
- **Six pairwise ortholog sets**
- **Six replicates of 3-quant-seq per species of basal leaflets from 2nd and 3rd leaf at 5 leaf stage**

	Accuracy	F1	R^2	Spearman's r
Washburn et al. 2019	**0.626**	**0.624**	-	-
AMII model	0.587	0.575	0.092	0.302
NRC model	0.599	0.591	**0.148**	**0.358**

Current model explains ~15% of variance in effect size

Pro-Seq and Quant-Seq data analysis



Correlation of ortholog gene expression ranks and counts

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amii